

MS101 – Makerspace
2023-24/II (Spring Semester)
EE Quiz - 1 Jan 31, 2024 (Wed)
Numerical Answers

Question	Numerical answers	Comments
1	a) 24 V b) 21.82 V c) 4.76 W	
2	$R_L = 90 \text{ k}\Omega$	
3	a) $i_1 = 150 \text{ mA}$, $i_2 = 150 \text{ mA}$, $i_3 = -350 \text{ mA}$, $i_4 = -350 \text{ mA}$ b) $V(A) = 5.5 \text{ V}$, $V(B) = 2.5 \text{ V}$, $V(C) = 13 \text{ V}$	
4	$I_d = 14.46 \text{ mA}$	
5	$v_{DM} = -128.2 \text{ mV}$ $v_{CM} = 5.0641 \text{ V}$ $v_{out} \approx -2.5 \text{ V}$	
6	Voltage gain = 1.5 Current gain = 1500 Power gain = 2250	
7	a) $V_{DUT} = 4.5 \text{ V}$, $V_{Out} = -3 \text{ V}$ b) $V_{DUT} = -7 \text{ V}$, $I_{DUT} = -2 \text{ mA}$	